

Dealing with the Inventory Risk: A Solution to the Market Making Problem

Stochastic Control Approach to Market Making
Project based on the paper by Guéant, Lehalle and Fernandez-Tapia

Aya El Moudden Aymen Imad Johanna Rakotovao

M2 Quantitative Finance – Stochastic Modelling/Control in Finance
Academic year 2025–2026

Introduction and Modelling

Motivation	Course link	Model setup	Handoff
Why inventory risk matters in market making	Optimal dealing under constraints	Variables, controls, execution and objective	Inputs for Bellman / HJB

Scope

This part introduces the economic problem and all core variables before the HJB derivation, the change of variables and the numerical implementation.

Why this paper?

Market making problem

A market maker continuously posts a bid price to buy and an ask price to sell. She earns the bid-ask spread, but she also carries inventory risk when executions are unbalanced.

Central question. How should bid and ask quotes be chosen when execution probability, spread revenue and inventory risk interact?

- Close quotes: faster executions, lower revenue per trade.
- Distant quotes: higher revenue per trade, lower execution probability.
- Inventory creates exposure to price movements.

Connection with the course

The paper is directly connected to the course chapter on **Optimal Dealing Under Constraints**.

Course concepts

- liquidity takers: market orders;
- liquidity providers: limit orders;
- market makers: liquidity provision under constraints.

In this paper

- bid/ask quotes are controls;
- executions are random;
- cash and inventory must be managed.

Course link

The paper studies this setting in a simplified stochastic control model: choosing optimal bid/ask quote distances under inventory limits.

Economic trade-off: spread vs. inventory risk

Source of revenue

- buy at the bid;
- sell at the ask;
- capture the bid-ask spread.

Source of risk

- executions arrive randomly;
- inventory becomes long or short;
- inventory value is exposed to price volatility.

Inventory risk

If the market maker is long and the price falls, she loses money. If she is short and the price rises, she also loses money.

Market state	Controls	Executions	Objective
S_t, q_t, X_t	δ_t^b, δ_t^a	N_t^b, N_t^a with intensities $Ae^{-k\delta}$	maximize CARA utility of $X_T + q_T S_T$

Idea

The market maker updates quote distances according to current inventory and market risk. Executions then update cash and inventory, which feed back into future decisions.

Reference price and quote-distance controls

The reference price, interpreted as a stylized mid-price or fair price, follows

$$dS_t = \sigma dW_t.$$

The market maker chooses **quote distances**, not a single fixed spread:

$$\delta_t^b = S_t - S_t^b, \quad \delta_t^a = S_t^a - S_t.$$

Equivalently,

$$S_t^b = S_t - \delta_t^b, \quad S_t^a = S_t + \delta_t^a.$$

Interpretation

Small distances increase execution probability. Larger distances improve the price obtained if execution occurs, but make execution less likely.

Executions: point processes and intensities

Executions are modelled by two point processes:

- N_t^b : executions at the bid, hence purchases by the market maker;
- N_t^a : executions at the ask, hence sales by the market maker.

Execution intensities decrease with quote distance:

$$\lambda^b(\delta^b) = Ae^{-k\delta^b}, \quad \lambda^a(\delta^a) = Ae^{-k\delta^a}.$$

Liquidity interpretation

A measures the baseline execution frequency; k measures how fast execution probability decreases when quotes move away from the reference price.

Inventory and cash dynamics

The inventory is the signed number of shares held by the market maker:

$$q_t = N_t^b - N_t^a.$$

A bid execution increases inventory; an ask execution decreases it.

Cash changes only when trades occur:

$$dX_t = (S_t + \delta_t^a)dN_t^a - (S_t - \delta_t^b)dN_t^b.$$

Inventory bounds

Inventory is restricted to $[-Q, Q]$. At $q = Q$ the market maker stops posting bid quotes; at $q = -Q$ she stops posting ask quotes. These bounds act as risk limits.

Objective: CARA utility of terminal P&L

Terminal mark-to-market wealth is

$$X_T + q_T S_T.$$

The paper uses CARA utility:

$$U(w) = -\exp(-\gamma w), \quad \gamma > 0.$$

The market maker solves

$$\sup_{(\delta_t^a), (\delta_t^b) \in \mathcal{A}} \mathbb{E}[-\exp(-\gamma(X_T + q_T S_T))].$$

Financial reading

The objective is not only to maximize spread revenue, but also to penalize risky terminal P&L generated by inventory exposure.

Notation summary before the HJB

Notation	Meaning	Economic role
S_t	reference price	stylized mid-price / fair price
S_t^b, S_t^a	bid and ask quotes	prices posted by the market maker
δ_t^b, δ_t^a	quote-distance controls	decision variables
N_t^b, N_t^a	execution counters	purchases and sales
q_t	signed inventory	long if positive, short if negative
X_t	cash account	cash generated by trades
A, k	liquidity parameters	level and decay of execution intensity
Q, γ	risk parameters	inventory limit and risk aversion

Modelling assumptions and limits

- The reference price is exogenous and follows an arithmetic Brownian motion.
- Order sizes are constant and normalized to one.
- Execution intensities are exponential in quote distance.
- Inventory is bounded between $-Q$ and Q .
- The baseline model ignores drift and market impact; these are discussed as extensions in the paper.

Why this is useful

The assumptions are stylized, but they make the inventory-risk mechanism clear and allow the HJB system to be reduced to linear ODEs in the next part.

Handoff to the HJB formulation

Inputs for the next part

The modelling layer has defined:

- state variables: time, cash, inventory and reference price;
- controls: bid and ask quote distances;
- random sources: Brownian price moves and random executions;
- objective: expected CARA utility of terminal mark-to-market P&L.

These elements are precisely the inputs of the Bellman equation and the HJB system developed next.

From Market Making to Stochastic Control

Goal of the market maker

A market maker continuously posts:

- a **bid price** to buy assets,
- an **ask price** to sell assets.

Its objective is to earn money from the bid-ask spread while controlling the risk generated by inventory fluctuations.

State variables

- S_t : reference (mid) price of the asset

$$dS_t = \sigma dW_t$$

- q_t : inventory held by the market maker

$$q_t > 0 \Rightarrow \text{long inventory}$$

$$q_t < 0 \Rightarrow \text{short inventory}$$

- X_t : cash process generated by executed trades

Control variables

The market maker controls the bid and ask quotes through:

$$\delta_t^b = S_t - S_t^b, \quad \delta_t^a = S_t^a - S_t$$

- Small δ : quote close to mid-price $\Rightarrow$ high execution probability
- Large δ : quote far from mid-price $\Rightarrow$ lower execution probability

Optimization problem

The objective is to maximize the expected utility of terminal wealth:

$$u(t, x, q, s) = \sup_{\delta^a, \delta^b} \mathbb{E} \left[-e^{-\gamma(X_T + q_T S_T)} \right]$$

The total mark-to-market value of the portfolio at time T is expressed by:

$$X_T + q_T S_T$$

Main trade-off

Spread Profit $\longleftrightarrow$ Inventory Risk

Bellman Principle and HJB Equation

Dynamic Programming Principle

At each instant, the market maker chooses optimal bid/ask quotes while taking into account:

- future price uncertainty,
- future executions,
- future inventory risk.

Bellman Principle

The optimal strategy at time t is obtained by combining:

- the immediate expected gain,
- the optimal future value function.

Value function

$$u(t, x, q, s) = \sup_{\delta^a, \delta^b} \mathbb{E} \left[-e^{-\gamma(X_T + q_T S_T)} \right]$$

The state variables are:

$$(t, x, q, s)$$

where t is current time, x is cash position, q is inventory, and s current asset price.

Hamilton-Jacobi-Bellman equation

Applying dynamic programming leads to the HJB equation:

$$\partial_t u + \frac{1}{2} \sigma^2 \partial_{ss} u + \sup_{\delta^b} (\cdots) + \sup_{\delta^a} (\cdots) = 0$$

Interpretation of each term

- $\partial_t u$ represents the evolution of the value function over time
- $\frac{1}{2} \sigma^2 \partial_{ss} u$ represents the diffusion term associated with Brownian price dynamics
- $\sup_{\delta^b}, \sup_{\delta^a}$ represents optimization over bid and ask quotes
- execution intensities depend on the chosen quotes:

$$\lambda(\delta) = A e^{-k\delta}$$

Main difficulty: the HJB equation is nonlinear and strongly coupled with the inventory process.

How can we solve this nonlinear HJB equation?

Key Idea

Introduce a clever change of variables exploiting:

- CARA exponential utility
- exponential arrival intensities

Nonlinear HJB $\implies$ Linear system of ODEs

Change of Variables

Main idea of the paper

The original HJB equation is nonlinear and difficult to solve directly.

The authors exploit the specific structure of:

- the CARA utility function,
- the exponential execution intensities.

Key Transformation

The value function is written under the form:

$$u(t, x, q, s) = -e^{-\gamma(x+qs)} v_q(t)^{-\gamma/k}$$

Interpretation of the decomposition

- $x + qs$ represents the mark-to-market wealth: Cash + Inventory Value
- $e^{-\gamma(x+qs)}$ comes from the exponential CARA utility: $U(x) = -e^{-\gamma x}$ The exponential form allows the dependence in (x, s) to factorize naturally.

- $v_q(t)$ captures the remaining effect of time, inventory dynamics, and execution probabilities.

Why is the exponential intensity important?

Execution intensities are modeled by:

$$\lambda(\delta) = Ae^{-k\delta}$$

Hence:

- quotes closer to the mid-price are executed faster,
- the exponential structure simplifies the Hamiltonian terms.

Fundamental Result

Thanks to this transformation:

$$\text{Nonlinear HJB PDE} \implies \text{Linear system of ODEs}$$

This is the central mathematical contribution of the paper.

Linear System of ODEs

After applying the change of variables, the nonlinear HJB equation becomes the following linear system:

$$\dot{v}_q = \alpha q^2 v_q - \eta(v_{q-1} + v_{q+1})$$

with

$$\alpha = \frac{k}{2}\gamma\sigma^2, \quad \eta = A\left(1 + \frac{\gamma}{k}\right)^{-(1+k/\gamma)}$$

Interpretation of the different terms

- The term $\alpha q^2 v_q$ corresponds to the inventory risk penalty.
When the inventory magnitude $|q|$ becomes large, the market maker is more exposed to price fluctuations since the inventory value $q_t S_t$ becomes more volatile.
The quadratic term q^2 therefore penalizes large inventory positions.
- The terms v_{q-1} and v_{q+1} represent inventory transitions.
Indeed:
 - a buy execution changes the inventory from q to $q + 1$,
 - a sell execution changes the inventory from q to $q - 1$.

The value function therefore depends on neighboring inventory states.

Matrix interpretation

The system can also be written under matrix form:

$$\dot{v} = Mv$$

where M is a tridiagonal matrix describing the transitions between inventory states.

Key Insight

The inventory dynamics behave similarly to a controlled Markov chain: each trade moves the market maker from one inventory state to another.

Optimal Quotes

Once the linear ODE system is solved, the optimal bid quote is obtained explicitly:

$$\delta^{b*}(t, q) = \frac{1}{k} \ln \left(\frac{v_q(t)}{v_{q+1}(t)} \right) + \frac{1}{\gamma} \ln \left(1 + \frac{\gamma}{k} \right)$$

The optimal ask quote has a similar expression:

$$\delta^{a*}(t, q) = \frac{1}{k} \ln \left(\frac{v_q(t)}{v_{q-1}(t)} \right) + \frac{1}{\gamma} \ln \left(1 + \frac{\gamma}{k} \right)$$

Interpretation of the formula

- The ratio $\frac{v_q}{v_{q+1}}$ measures the marginal value of increasing inventory.
- The logarithmic term adjusts the quote depending on the current inventory level.
- The term $\frac{1}{\gamma} \ln \left(1 + \frac{\gamma}{k} \right)$ corresponds to the profitability/risk trade-off.

Key Financial Intuition

The quotes are dynamically skewed in order to control inventory risk.

Case 1: Positive inventory ($q > 0$)

The market maker already holds too many shares.

- He wants to reduce inventory exposure.
- The ask quote becomes more aggressive:
 $\delta^{a*} \downarrow$
so selling becomes easier.
- The bid quote becomes less attractive:
 $\delta^{b*} \uparrow$
in order to avoid buying additional shares.

Case 2: Negative inventory ($q < 0$)

The market maker is short and wants to buy back shares.

- Bid quotes become more aggressive.
- Ask quotes become less aggressive.

Main Idea

Optimal quotes are not symmetric: they continuously adapt in order to bring inventory back toward zero.

Asymptotic Behaviour

When the horizon T becomes large:

$$\delta^{b*}(t, q) \longrightarrow \delta_{\infty}^{b*}(q)$$

Key result

The quotes converge toward a stationary regime related to:

an eigenvalue problem

Interpretation

Far from terminal time, the strategy becomes almost time-independent.

Closed-Form Approximation

The asymptotic optimal bid quote is approximated by:

$$\delta_{\infty}^{b*}(q) \simeq \frac{1}{\gamma} \ln \left(1 + \frac{\gamma}{k} \right) + \frac{2q+1}{2} \sqrt{\frac{\sigma^2 \gamma}{2kA(1 + \gamma/k)^{1+k/\gamma}}}$$

Main insight

Optimal quotes depend explicitly on:

- volatility σ
- liquidity parameter A
- risk aversion γ
- inventory level q

Economic Interpretation of Parameters

Effect of volatility σ

$$\sigma \uparrow \Rightarrow \text{spread} \uparrow$$

Higher volatility increases inventory risk.

Effect of liquidity A

$$A \uparrow \Rightarrow \text{spread} \downarrow$$

More liquidity reduces execution risk.

Effect of risk aversion γ

$$\gamma \uparrow \Rightarrow \text{more conservative quotes}$$

Theoretical Takeaways

- Market making can be formulated as a stochastic control problem
- The HJB equation is initially nonlinear and difficult to solve
- The paper introduces a remarkable change of variables
- This transforms the problem into a linear ODE system
- Explicit approximations provide strong financial intuition

From the HJB formulation to numerical implementation

Objective:

The objective of this part is to implement numerically the Avellaneda–Stoikov model in order to simulate the behavior of a market maker exposed to inventory risk.

Model Principle:

- The market maker continuously posts:
 - a bid quote (buy),
 - an ask quote (sell).
- The quotes are computed in order to:
 - maximize profits,
 - control inventory risk.
- The market price follows a Brownian motion:

$$dS_t = \sigma dW_t$$

Model Parameters

Parameter	Role	Economic Interpretation
σ	Volatility	Measures market price fluctuations
γ	Risk aversion	Controls the market maker's sensitivity to inventory risk
A	Base intensity	Average order arrival intensity
k	Liquidity parameter	Controls the decay of executions with distance to the mid-price
q	Inventory	Number of assets held by the market maker
T	Time horizon	End of the trading period

Market Maker Objective:

$$\max \mathbb{E} \left[-e^{-\gamma(X_T + q_T S_T)} \right]$$

Assumptions and Simplifications

Main Assumptions:

- Order arrivals follow Poisson processes
- Execution intensities are exponential:

$$\lambda(\delta) = Ae^{-k\delta}$$

- The market maker can continuously adjust quotes

Simplifications Used:

- No real market impact
- No transaction costs
- Constant market liquidity
- Stationary parameters

Purpose of These Simplifications:

To obtain a numerically stable model while preserving the main economic mechanisms of market making.

Numerical Simulation of the Model

Numerical Approach

The Avellaneda–Stoikov model is implemented numerically in order to compute the optimal bid and ask quotes of the market maker under inventory risk.

Simulation Pipeline

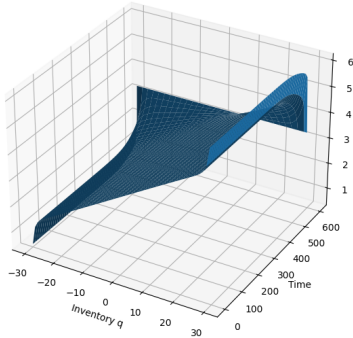
- 1 Simulation of the market price process
- 2 Construction of the tridiagonal matrix M
- 3 Numerical resolution of the ODE system

$$v(t) = e^{-M(T-t)}1$$

- 4 Computation of optimal quotes
- 5 Simulation of order executions
- 6 Update of:
 - inventory,
 - cash process,
 - P&L.

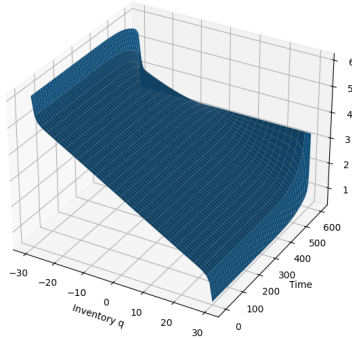
Optimal Quotes and Spread Surfaces

Optimal Bid Quotes



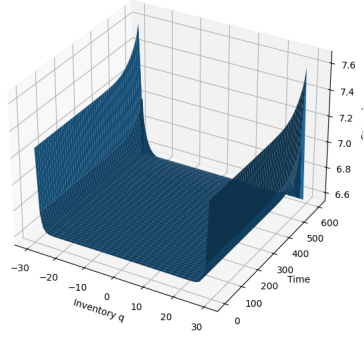
Optimal Bid Quotes

Optimal Ask Quotes



Optimal Ask Quotes

Optimal Spread

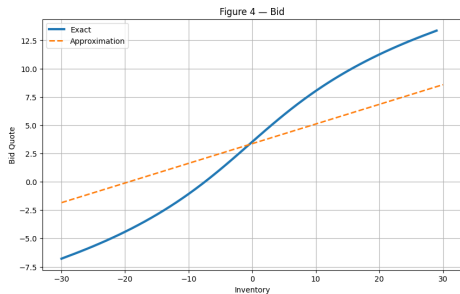


Optimal Spread

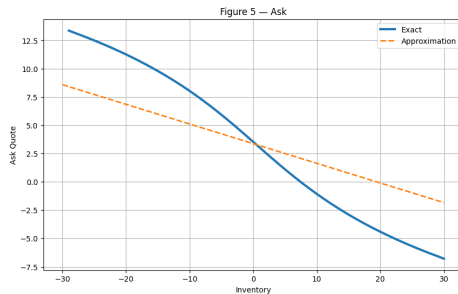
Analysis and Interpretation

- The bid and ask quotes dynamically depend on both time and inventory.
- When the inventory becomes strongly positive, the market maker reduces bid aggressiveness and increases selling incentives.
- Conversely, when the inventory is negative, the strategy encourages buying activity.
- The spread widens for extreme inventory values, reflecting stronger inventory risk control.
- These numerical results are fully consistent with the theoretical behavior predicted by the Avellaneda–Stoikov model.

Exact vs Approximate Quotes



Exact vs Approximate Bid Quote



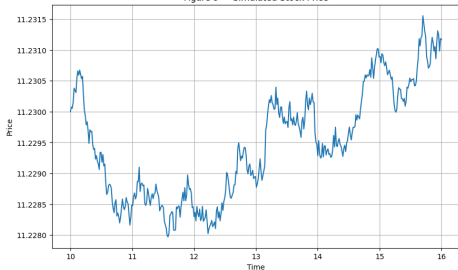
Exact vs Approximate Ask Quote

Analysis and Interpretation

- The asymptotic approximation follows the global trend of the exact solution.
- Differences become larger for extreme inventory values.
- The exact solution captures stronger inventory risk effects.
- The approximation remains accurate for moderate inventories.

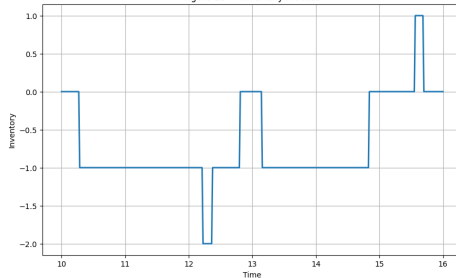
Price Dynamics and Inventory Process

Figure 9 — Simulated Stock Price



Simulated Stock Price

Figure 10 — Inventory Process



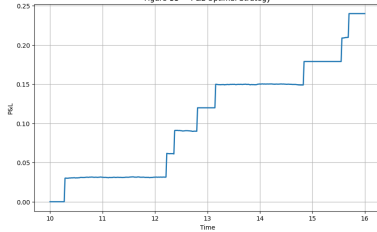
Inventory Process

Analysis and Interpretation

- The stock price follows a stochastic Brownian-type evolution with random fluctuations.
- The inventory changes according to bid and ask order executions.
- Inventory remains globally close to zero, showing effective inventory risk control.
- The strategy dynamically adjusts positions to avoid excessive exposure.

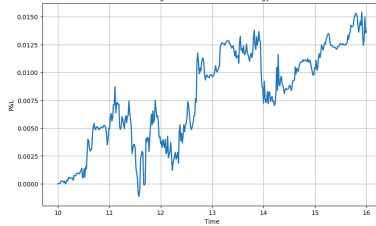
P&L Comparison of Trading Strategies

Figure 11 — P&L Optimal Strategy



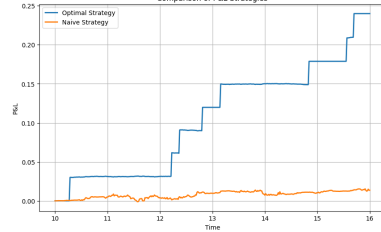
Optimal Strategy

Figure 12 — P&L Naive Strategy



Naive Strategy

Comparison of P&L Strategies



Strategy Comparison

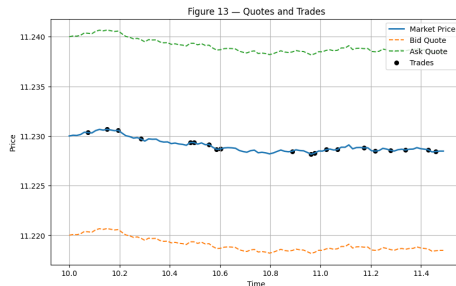
Analysis and Interpretation

- The optimal strategy produces a higher and more stable P&L.
- The naive strategy generates lower profits and stronger fluctuations.
- Inventory-aware quotes improve both profitability and risk control.
- These results confirm the efficiency of the Avellaneda–Stoikov framework.

Quotes and Executed Trades

Analysis

- Bid and ask quotes evolve dynamically with the market price.
- Executed trades occur close to the optimal quotes.
- The spread remains positive during the simulation.
- The strategy continuously adapts to inventory variations.



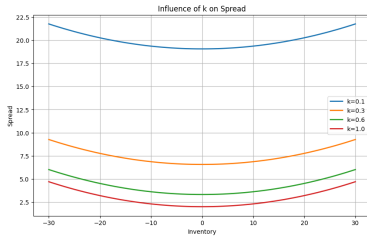
Market Price, Quotes and Executed Trades

Interpretation

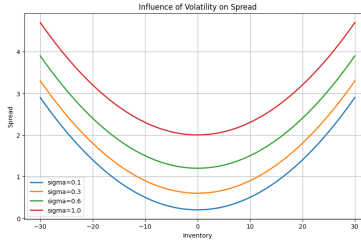
The model generates realistic bid and ask quotes while maintaining effective inventory risk control and stable trading behavior.

Metric	Value
MAE Bid	3.30
RMSE Bid	3.60
Max Error	4.95

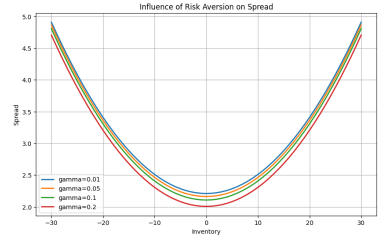
Sensitivity Analysis of the Spread



Influence of k



Influence of σ



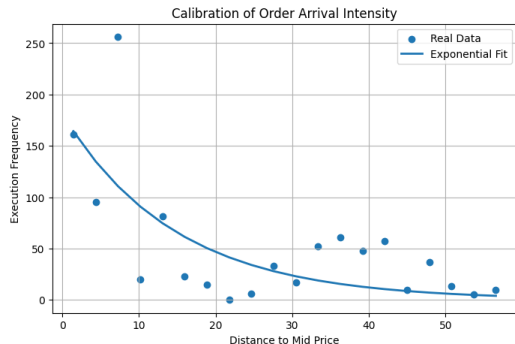
Influence of γ

Analysis

- Higher risk aversion strengthens inventory protection.
- The spread remains minimal near zero inventory.

Parameter	Effect on Spread
$\sigma \uparrow$	Spread increases
$k \uparrow$	Spread decreases
$\gamma \uparrow$	Stronger inventory control

Calibration Using Real Market Data



Calibration of Order Arrival Intensity Using
Coinbase Data

Analysis

- Real BTC-USD market data collected from Coinbase

Parameter	Value
σ_{real}	1.20×10^{-5}
A	180.96
k	0.0678
Bid	78202.54
Ask	78202.55

Interpretation

The calibration shows that the Avellaneda–Stoikov model captures realistic market execution dynamics using real Coinbase data.

Conclusion

- The Avellaneda–Stoikov model provides an efficient framework for market making under inventory risk.
- The project reproduces the main mechanisms of the original model.
- Simulated market data were used for the numerical implementation.
- The implementation remains a simplified version of the original article.
- The numerical results remain globally coherent with the theoretical model.
- The project highlights the relevance of stochastic control methods in quantitative finance.